

DPP-6

- Q.1 For non-zero vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $|\vec{a} \times \vec{b} \cdot \vec{c}| = |\vec{a}| |\vec{b}| |\vec{c}|$ holds if and only if;
- (A) $\vec{a} \cdot \vec{b} = 0$, $\vec{b} \cdot \vec{c} = 0$ (B) $\vec{c} \cdot \vec{a} = 0$, $\vec{a} \cdot \vec{b} = 0$
(C) $\vec{a} \cdot \vec{c} = 0$, $\vec{b} \cdot \vec{c} = 0$ (D) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$
- Q.2 The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ & $\vec{c} = 3\hat{i} + \hat{j} + 4\hat{k}$ are so placed that the end point of one vector is the starting point of the next vector. Then the vectors are
- (A) not coplanar (B) coplanar but cannot form a triangle
(C) coplanar but can form a triangle (D) coplanar & can form a right angled triangle
- Q.3 Given the vectors
- $$\vec{u} = 2\hat{i} - \hat{j} - \hat{k}$$
- $$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$$
- $$\vec{w} = \hat{i} - \hat{k}$$
- If the volume of the parallelepiped having $-c\vec{u}$, $\vec{v}$ and $c\vec{w}$ as concurrent edges, is 8 then 'c' can be equal to
- (A) ± 2 (B) 4 (C) 8 (D) can not be determined
- Q.4 Given $\vec{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + 2\hat{j}$; $(\vec{a} \wedge \vec{b}) = \pi/2$, $\vec{a} \cdot \vec{c} = 4$ then
- (A) $[\vec{a} \vec{b} \vec{c}]^2 = |\vec{a}|$ (B) $[\vec{a} \vec{b} \vec{c}] = |\vec{a}|$ (C) $[\vec{a} \vec{b} \vec{c}] = 0$ (D) $[\vec{a} \vec{b} \vec{c}] = |\vec{a}|^2$
- Q.5 The set of values of m for which the vectors $\hat{i} + \hat{j} + m\hat{k}$, $\hat{i} + \hat{j} + (m+1)\hat{k}$ & $(\hat{i} - \hat{j} + m\hat{k})$ are non-coplanar:
- (A) R (B) $R - \{1\}$ (C) $R - \{-2\}$ (D) ϕ
- Q.6 Let a, b, c be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ & $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then c is:
- (A) the A.M. of a & b (B) the G. M. of a & b
(C) the H. M. of a & b (D) equal to zero.
- Q.7 Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ & $\vec{b}$. If the angle between $\vec{a}$ & $\vec{b}$ is $\frac{\pi}{6}$ then $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 =$
- (A) 0 (B) 1
(C) $\frac{1}{4} (a_1^2 + a_2^2 + a_3^2) (b_1^2 + b_2^2 + b_3^2)$ (D) $\frac{3}{4} (a_1^2 + a_2^2 + a_3^2) (b_1^2 + b_2^2 + b_3^2) (c_1^2 + c_2^2 + c_3^2)$
- Q.8 For three vectors $\vec{u}$, $\vec{v}$, $\vec{w}$ which of the following expressions is not equal to any of the remaining three?
- (A) $\vec{u} \cdot (\vec{v} \times \vec{w})$ (B) $(\vec{v} \times \vec{w}) \cdot \vec{u}$ (C) $\vec{v} \cdot (\vec{u} \times \vec{w})$ (D) $(\vec{u} \times \vec{v}) \cdot \vec{w}$

- Q.9 The vector $\vec{c}$ is perpendicular to the vectors $\vec{a} = (2, -3, 1)$, $\vec{b} = (1, -2, 3)$ and satisfies the condition $\vec{c} \cdot (\hat{i} + 2\hat{j} - 7\hat{k}) = 10$. Then the vector $\vec{c} =$
 (A) $(7, 5, 1)$ (B) $(-7, -5, -1)$ (C) $(1, 1, -1)$ (D) none
- Q.10 Let $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$ & $\vec{c} = \alpha\vec{a} + \beta\vec{b}$. If the vectors, $\hat{i} - 2\hat{j} + \hat{k}$, $3\hat{i} + 2\hat{j} - \hat{k}$ & $\vec{c}$ are coplanar then $\frac{\alpha}{\beta}$ is :
 (A) 1 (B) 2 (C) 3 (D) -3
- Q.11 A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ radians/sec. If $\vec{\omega}$ points in the direction of $\hat{i} + \hat{j} + \hat{k}$ then the equation to the locus of the points having tangential speed 20 m/sec. is :
 (A) $x^2 + y^2 + z^2 - xy - yz - zx - 1 = 0$
 (B) $x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 1 = 0$
 (C) $x^2 + y^2 + z^2 - xy - yz - zx - 2 = 0$
 (D) $x^2 + y^2 + z^2 - 2xy - 2yz - 2zx - 2 = 0$
- Q.12 A rigid body rotates with constant angular velocity ω about the line whose vector equation is, $\vec{r} = \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$. The speed of the particle at the instant it passes through the point with p.v. $2\hat{i} + 3\hat{j} + 5\hat{k}$ is :
 (A) $\omega\sqrt{2}$ (B) 2ω (C) $\omega/\sqrt{2}$ (D) none
- Q.13 Given 3 vectors
 $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$; $\vec{V}_2 = b\hat{i} + c\hat{j} + a\hat{k}$; $\vec{V}_3 = c\hat{i} + a\hat{j} + b\hat{k}$
 In which one of the following conditions $\vec{V}_1$, $\vec{V}_2$ and $\vec{V}_3$ are linearly independent?
 (A) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
 (B) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$
 (C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$
 (D) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$
- Q.14 If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ & $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, then the vector $\vec{c}$ such that $\vec{a} \cdot \vec{c} = 2$ & $\vec{a} \times \vec{c} = \vec{b}$ is
 (A) $\frac{1}{3}(3\hat{i} - 2\hat{j} + 5\hat{k})$ (B) $\frac{1}{3}(-\hat{i} + 2\hat{j} + 5\hat{k})$ (C) $\frac{1}{3}(\hat{i} + 2\hat{j} - 5\hat{k})$ (D) $\frac{1}{3}(3\hat{i} + 2\hat{j} + \hat{k})$

One or more than one is/are correct:

- Q.15 If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ & $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?
 (A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs (B) $[\vec{a} \vec{b} \vec{c}] = |\vec{b}|$
 (C) $[\vec{a} \vec{b} \vec{c}] = |\vec{c}|^2$ (D) $|\vec{b}| = |\vec{c}|$